THE KINETICS OF HYDROGEN PEROXIDE DECOMPOSITION

(PART I)

OBJECTIVE

- To determine the kinetic parameters (rate, order, rate constant) of a reaction using the method of initial rates

INTRODUCTION

Studying the kinetics of a reaction is the key to understanding in detail how the reaction actually occurs—in other words, the *mechanism* of the reaction. In a kinetic study, one investigates the rates at which reactions occur and how those rates are affected by changes in the temperature, pressure, concentration, and the presence of catalysts. Once these factors are understood, conditions may be optimized to speed up or slow down a reaction as desired. In addition, information about reaction intermediates and electron movement during the reaction can be gained. A discussion of reaction rates, rate constants, laws and the theory of chemical kinetics may be found in your textbook.

The speed of a reaction is of practical interest, particularly in a commercial production setting. The speed—or rate—is usually described in the following way:

$$rate = \frac{change\ in\ concentration}{elapsed\ time} = \frac{\Delta[\ ]}{\Delta t}$$

where [ ] signifies the concentration in molarity ($moles/L$) of the compound you are observing. In this experiment, the quantity of O_2 gas produced will be monitored (instead of molarity) as a function of the elapsed time. The rate of reaction can then be written.

$$rate = \frac{\Delta(moles\ of\ O_2)}{\Delta t} = \frac{\Delta(O_2\ pressure)}{\Delta t}$$

(Recall $PV = nRT$ where the volume of the reaction chamber and T are held constant. Rate units are atm/min or $mm\ Hg/min$, depending on what pressure units are used.)

In this experiment you will investigate the kinetics of the reaction:

$$2\ H_2O_{2\,(g)} \rightarrow O_{2\,(g)} + 2\ H_2O_{(l)} \qquad (1)$$

Hydrogen peroxide, which contains oxygen in the oxidation state of -1, can act as both an oxidizing agent and as a reducing agent. H_2O_2 is unstable and disproportionates (meaning: can be oxidized and reduced in the same reaction) according to the half reactions:

$$H_2O_{2\,(aq)} + H_2O_{(l)} \rightarrow \frac{3}{2}O_{2\,(g)} + 4\,H^+ + 4\,e^- \quad (2)$$

$$H_2O_{2\,(aq)} + 2\,H^+ + 2\,e^- \rightarrow 2\,H_2O_{(l)} \quad (3)$$

While the conversion of H_2O_2 to O_2 and H_2O is normally slow, adding iodide as a catalyst will increase the rate of reaction. In a reaction equation, catalysts are shown above the arrow:

$$2\,H_2O_{2\,(g)} \xrightarrow{I^-} 2\,H_2O_{(l)} + O_{2\,(g)} \quad (4)$$

In this case, the I^- ion is considered a *homogeneous catalyst* because it is dissolved in the solution. While I^- affects the speed of the reaction and is involved in the rate-determining (or slow) step of the reaction, it is not part of the overall balanced equation because it is regenerated in the last step.

Reaction Order

In Week 1, we will determine as quantitatively as possible the relationships between the concentrations of H_2O_2 and $I^-$and the rate of the reaction, with the goal of determining the rate law of the reaction:

$$rate = k[H_2O_2]^x[I^-]^y \quad (5)$$

Your job today is to **DETERMINE THE EXPONENTS x AND y** in the rate law (equation 5) using the method of initial rates. (See your textbook for additional information on how to do this). Once you have determined x and y, the value of the rate constant $\boldsymbol{k}$ can be calculated.

PRE-LAB: BEFORE YOU COME TO LAB THIS WEEK PLEASE FILL OUT THE FOLLOWING TABLES IN YOUR NOTEBOOK

Table 1. Solution Components for Mixtures, 1, 2 and 3.

Reagents for solutions	**Mixture 1**	**Mixture 2**	**Mixture 3**	**Volume for Mixtures**
0.10 M KI	15.0 mL	15.0 mL	30.0 mL	Initial volume
$Distilled\ water$	35.0 mL	25.0 mL	20.0 mL	Initial volume
3% H_2O_2 (0.88 M)	10.0 mL	20.0 mL	10.0 mL	Initial volume
	60.0 mL	**60.0 mL**	**60.0 mL**	**Total volume**

Use $M_1V_1 = M_2V_2$ to calculate the final concentrations of H_2O_2 and I^-in each of your mixtures:

Table 2. Final $[H_2O_2]$and $[I^-]$ in mixtures, 1, 2 and 3

Component	Mixture 1	Mixture 2	Mixture 3
$[\boldsymbol{H_2O_2}]$			
$[\boldsymbol{KI}] = [\boldsymbol{I^-}]$			

In this lab you will use the *method of initial rates* in order to determine the reaction orders (exponents x and y) with respect to H_2O_2 and I^-.

General Process:

1. Measure the average rate of the reaction (Mixture 1)
2. Vary the concentration of only one reactant (Mixture 2) - measure the average rate again
3. Vary the concentration of the other reactant (Mixture 3) - measure the average rate again
4. Ask yourself: How did changing the concentration of each reactant while holding the other reactant constant effect the average rate of the reaction?

PROCEDURE

A. Concentration Effects on reaction rate

1. Set up the apparatus illustrated in Figure 1. Place a stir bar in the 250 mL Erlenmeyer flask that will contain your reaction mixtures. The connections will need to be airtight under low pressure. Watch for signs of gas leaks as you take measurements.

2. Clamp a thermometer so it measures the temperature of the water bath. The water in the bath should be at room temperature ($20℃ - 24℃$) through ALL three reactions. Record that actual temperature in your lab notebook.

3. Carefully measure the quantities prescribed for Mixture 1 given in Table 1 (page 2). Add the KI and distilled water to a 250 mL Erlenmeyer reaction flask. Add a stir bar to the water bath.

4. Measure the prescribed volume of H_2O_2 into a graduated cylinder and set aside.

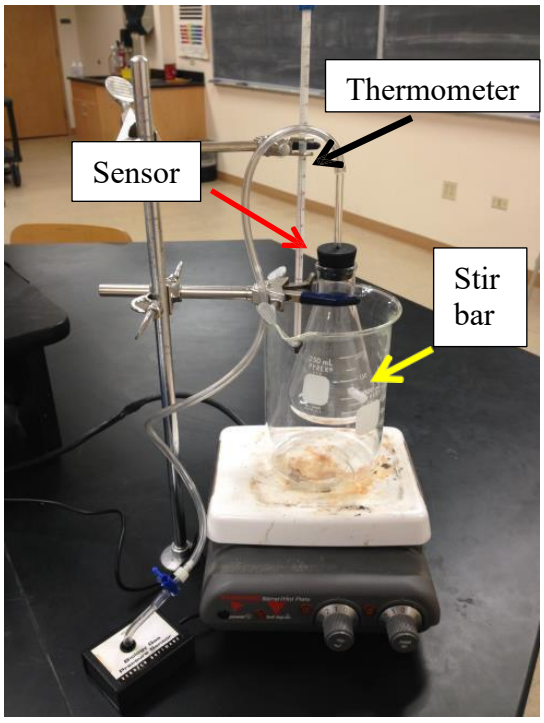


Figure 1. Apparatus set-up.

5. Start stirring at a moderate speed and maintain this rate throughout each experimental run.

6. Plug in your pressure sensor and open Logger Pro. Check that an appropriate lab pressure reading is displayed in the upper left corner. If not, consult your instructor!

7. Adjust your Data Collection Parameters appropriately with pressure measured in atmospheres (atm) and time measured in minutes (min).

8. When all is ready, select COLLECT, quickly add the H_2O_2 to the reaction flask as completely as you can, insert the stopper firmly and twist so it is gas tight.

9. Continue to collect data for 8 to 10 minutes or until the stopper pops out of the Erlenmeyer flask. Select STOP to terminate data collection.

10. To analyze your data, select EXAMINE under the ANALYZE menu. Click and drag over the area of interest in your data - the most linear portion - to highlight it. This area will be 2-4 minutes long. Then choose Linear Fit under the ANALYZE menu. The equation for your selected data will appear on the data screen. Record this equation, paying particular attention to the slope which is equal to the average rate during your highlighted time period. The slope is change in pressure over change in time and is equal to the *rate* in the **rate law equation**.

$$rate = k[H_2O_2]^x[I^-]^y$$

11. Dispose of products in the waste container.

12. Repeat the data collection and analysis (steps 1-9) for Mixture 2 and Mixture 3 (from Table 1).

13. Select COLLECT to begin logging the next data set. When you select COLLECT a dialogue box will appear. Select STORE LATEST RUN.

14. These three mixture plots should be included it in your data section of your submitted lab notebook.

CALCULATIONS

1. For each mixture use the following steps to determine the rate constant, **k** from the **rate law equation**:

$$\boldsymbol{rate} = \boldsymbol{k}[H_2O_2]^x[I^-]^y$$

where, **rate** is the $\frac{\Delta concentration}{\Delta time}$, *but here is measured as the* $\frac{\Delta pressure\ (atm)}{\Delta time\ (min)}$

k is the rate constant. Think about units here as it must equal the units on the rate and cancel the concentrations (in M) of the reactants once you know x and y.

x is the reaction order with respect to hydrogen peroxide and **y** is the reaction order with respect to I^-.

2. Use the concentration of $[H_2O_2]$ and $[KI] = [I^-]$ for each mixture that you calculated in Table 2 for Pre-lab for the following calculations:
   a. Using the method of average rates (examples are given in your textbook for this process using the method of initial rates) determine the value of the exponents x and y.
   b. Using the rates (slope from linear line fit), concentrations and the exponents that you have determined calculate k for each mixture.
   c. Calculate an average k for Mixtures 1, 2, and 3. Report this value at the temperature (°C) of the water bath.
   d. Clean up your set up and be prepared to do Week 2 next week.

INCLUDE THE FOLLOWING IN YOUR CONCLUSION

1. Write a conclusion that reports the observed **orders of reaction** for H_2O_2 and I^-, the **average *k*** at your measured temperature for the decomposition of H_2O_2.

2. Include in your conclusion thoughts you have on what the order with respect to each reactant values tell us about the decomposition of H_2O_2.